

BLUESTOCK

Mutual Fund Analytics Platform

An End-to-End Financial Data Analytics Project

FINAL PROJECT REPORT

Prepared By

Darling — Project Team, Kallam Haranadhareddy Institute of Technology (KHIT)

Technologies Used

Python | Pandas | SQL | SQLite | Power BI | Jupyter Notebook

July 2026

Table of Contents

1. Executive Summary	3
2. Project Objectives	4
3. Project Architecture	5
4. Data Engineering Process	6
5. Exploratory Data Analysis	9
6. Fund Performance Analytics	12
7. Advanced Analytics	15
8. Power BI Dashboard	18
9. Key Findings	21
10. Conclusion	23

1. Executive Summary

The mutual fund industry in India generates an enormous volume of data every single day — NAV updates, SIP transactions, redemptions, folio records, and fund house disclosures. Making sense of this data manually is slow and error-prone, which is exactly the gap this project set out to close.

The Bluestock Mutual Fund Analytics Platform is an end-to-end analytics solution built to take raw mutual fund data and turn it into insights that are actually useful — for comparing fund performance, understanding investor behaviour, measuring risk, and spotting trends across the industry.

The project was built as a complete pipeline rather than a one-off analysis. It starts with raw CSV files, moves through a Python-based ETL process, lands in a structured SQLite database, and finally surfaces through an interactive Power BI dashboard. Every stage was designed to feed cleanly into the next, so the final dashboard reflects data that has already been validated, deduplicated, and organised.

Along the way, the project also goes beyond basic reporting. Risk-adjusted performance metrics such as the Sharpe and Sortino ratios, downside-risk measures like Value at Risk (VaR) and Conditional VaR, and behavioural analysis such as SIP continuity and investor cohorts were all built to give a more rounded picture of how funds and investors actually behave — not just how NAV moves day to day.

This report documents the complete workflow: how the data was engineered, what analysis was performed, what the dashboard shows, and what conclusions came out of the exercise. It is meant to be read by anyone reviewing the project for academic evaluation, and by anyone curious about how a mutual fund analytics platform is put together from scratch.

Project Snapshot

Aspect	Detail
Project Type	End-to-end financial data analytics platform
Core Technologies	Python, Pandas, SQL, SQLite, Power BI, Jupyter Notebook
Data Scope	Mutual fund NAV, AUM, transactions and investor records, 2022 – 2026
Database Design	Star schema with dimension and fact tables
Dashboard	4-page interactive Power BI report with slicers and drill-through
Key Outputs	Fund scorecard, risk metrics, investor segmentation, recommendation logic

2. Project Objectives

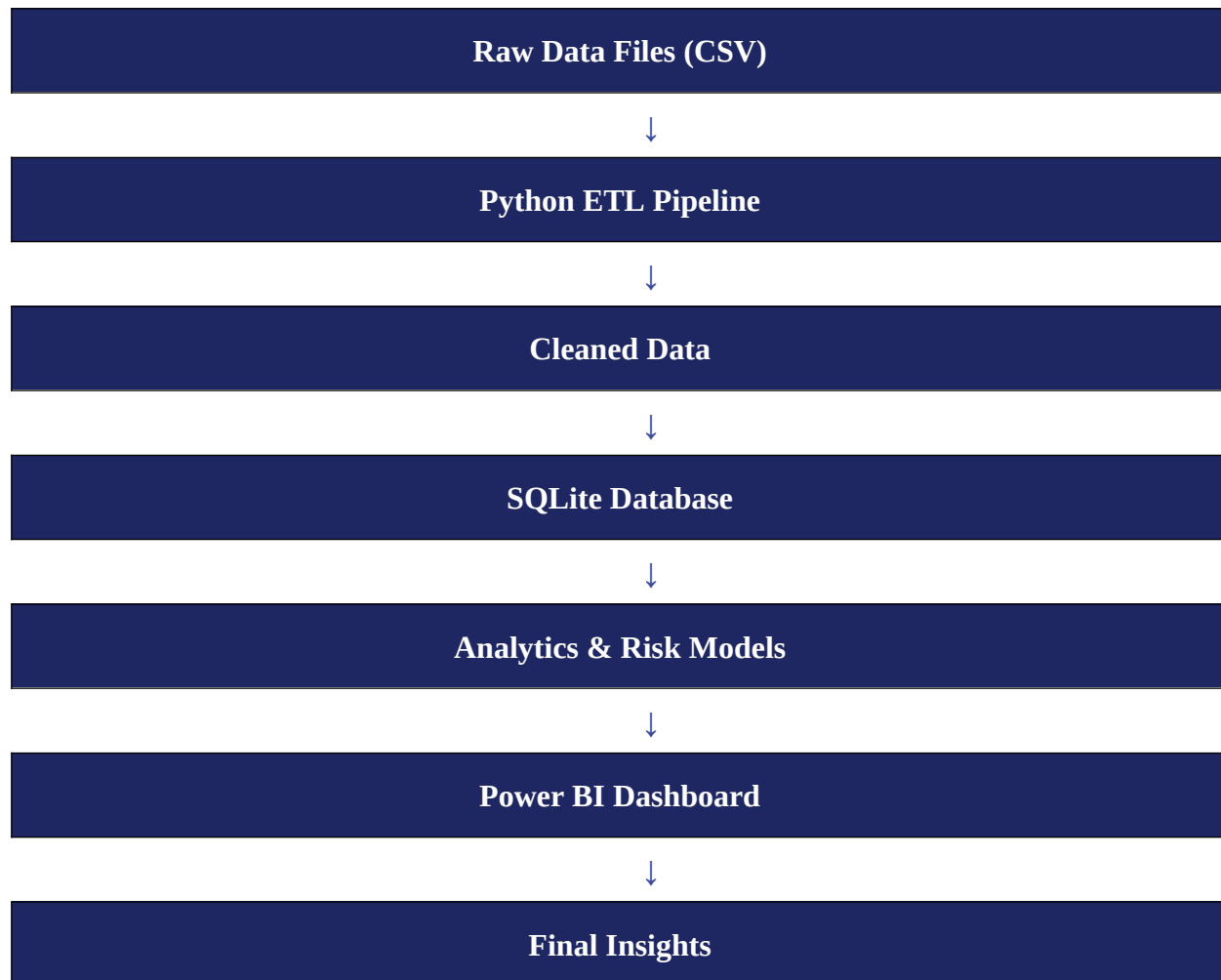
Before any code was written, the project was scoped around a clear set of goals. These objectives guided every design decision, from how the database was structured to what went on the final dashboard.

1. Build a reliable ETL pipeline that can take raw mutual fund CSV files and turn them into clean, analysis-ready datasets.
2. Clean and validate the underlying financial data — handling missing values, duplicate entries, inconsistent date formats, and outliers.
3. Store the processed data in a structured SQLite database that supports fast, repeatable queries.
4. Perform exploratory data analysis to understand NAV movement, AUM growth, and investor patterns before jumping into modelling.
5. Calculate standard investment performance metrics (CAGR, Sharpe, Sortino, Alpha, Beta, Maximum Drawdown) for every scheme in the dataset.
6. Analyse risk factors in more depth using Value at Risk, Conditional VaR, and rolling risk-adjusted return measures.
7. Understand investor behaviour — SIP patterns, cohort retention, geographic spread, and age-group preferences.
8. Build an interactive Power BI dashboard that lets a reviewer explore the data without needing to touch the underlying code.
9. Generate actionable insights that summarise what the data actually says about fund performance and investor trends.

Together, these objectives cover the full lifecycle of a data analytics project — not just the analysis itself, but the engineering work that has to happen before any chart can be trusted.

3. Project Architecture

The platform follows a linear, layered architecture. Each layer has a single responsibility, which keeps the pipeline easy to debug and easy to extend. If a new fund house's data needs to be added, for example, it only has to pass through the ETL layer — nothing downstream needs to change.



How the layers work together

Raw data enters the system as a set of CSV exports covering fund master data, daily NAV, AUM by fund house, and investor transactions. The Python ETL pipeline reads these files, applies the cleaning rules described in Section 4, and writes the results into the SQLite database using a star schema.

Once the data is in the database, a separate analytics layer runs on top of it — this is where performance metrics, risk models, and behavioural analysis are computed using Python and SQL queries. The results of this layer feed directly into Power BI, which connects to the same database and refreshes its visuals whenever the underlying tables are updated.

This separation between engineering, analytics and visualisation is intentional. It means the dashboard never has to deal with messy data directly, and the analytics scripts never have to worry about how something is being displayed — they just need to produce clean output tables.

4. Data Engineering Process

The project started with a handful of raw CSV datasets — one for fund master details, one for daily NAV history, one for AUM by fund house, and one for investor-level transactions. None of these files were analysis-ready on their own, so the first real chunk of work went into cleaning and organising them.

4.1 Cleaning and Validation

Python and Pandas handled the bulk of this work. The cleaning process was not a single script but a series of checks applied consistently across every dataset:

- Data loading — reading each CSV with the correct data types, rather than letting Pandas infer them and get dates or fund codes wrong.
- Missing value handling — some NAV entries and investor fields were blank; these were either filled using forward-fill logic (for NAV series) or flagged and excluded from downstream metric calculations.
- Duplicate removal — a small number of transaction records appeared more than once due to overlapping export windows; these were identified using a composite key of investor ID, fund code and transaction date.
- Date formatting — dates arrived in at least three different formats across files and were standardised to a single ISO format before loading.
- Data validation — NAV values were checked for negative numbers, AUM figures were checked against plausible ranges, and fund codes were cross-referenced against the master list to catch typos.
- Data quality checks — a final pass compared row counts before and after cleaning, and logged what was dropped and why, so nothing was silently discarded.

Once a dataset passed these checks, it was saved separately as a cleaned CSV before being loaded into the database. Keeping the cleaned files separate from the raw files made it much easier to re-run the load step without repeating the cleaning logic every time.

4.2 Database Design

The cleaned data was loaded into SQLite using a star schema — a small number of dimension tables describing 'what' and 'when', and a larger set of fact tables holding the actual measurements. This structure keeps queries fast and makes it straightforward to join fund details with performance numbers without repeating fund metadata in every row.

Dimension Tables

Table	Purpose
dim_fund	One row per mutual fund scheme — fund name, category, fund house, launch date, and expense ratio.
dim_date	A standard date dimension used to join NAV, transaction and AUM records to calendar attributes such as month, quarter and year.

Fact Tables

Table	Purpose
fact_nav	Daily NAV values for every scheme, used for return and volatility calculations.
fact_transactions	Investor-level SIP, lumpsum and redemption transactions.
fact_performance	Pre-computed performance metrics (CAGR, Sharpe, Sortino, Alpha, Beta) refreshed on each analytics run.
fact_aum	Monthly AUM by fund and fund house, used for growth and concentration analysis.

Every fact table links back to `dim_fund` through a fund code and to `dim_date` through a date key. This is a fairly standard warehousing pattern, but it made a real difference here — most of the analytics queries in later sections are joins across two or three of these tables, and the star schema keeps those joins simple and fast even as the dataset grows.

4.3 Why This Structure Was Chosen

An alternative would have been to keep everything in one wide table. That was tried early on and dropped quickly — it made the transaction table enormous, repeated fund metadata millions of times over, and made it awkward to add a new fact table later (for instance, `fact_aum` was added after the fact, and the star schema let it slot in without touching the other tables).

5. Exploratory Data Analysis

Before building any performance models, it was important to just look at the data — plot NAV trends, chart AUM growth, and get a feel for how investors were actually behaving. This step surfaced a few data quality issues that the cleaning process had missed, and it also shaped which metrics were worth calculating later.

5.1 NAV Trend Analysis

Daily NAV movement was studied across multiple schemes from 2022 to 2026 to understand overall market behaviour and to spot periods of unusual volatility. Plotting NAV over time for a sample of large-cap, mid-cap and debt schemes made it clear how differently these categories responded to the same market events — large-cap funds tracked the broader market fairly closely, while some mid-cap schemes showed sharper swings during correction periods.

[Insert NAV Trend Chart]

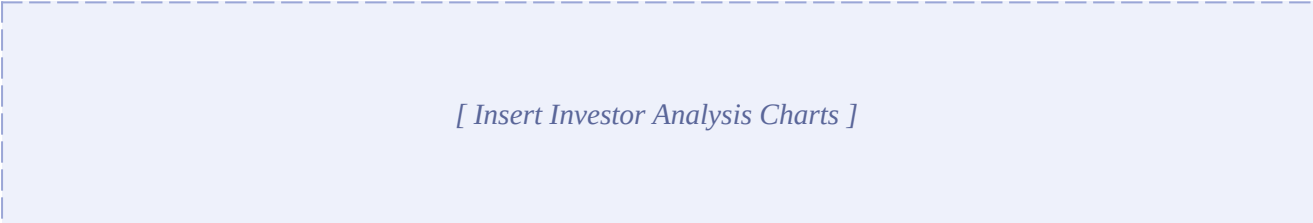
5.2 AUM Growth Analysis

Fund house growth was compared over the same period to identify which players were driving the bulk of industry AUM. A small number of large fund houses accounted for a disproportionate share of total assets, while a longer tail of smaller houses grew more slowly but showed steadier month-on-month inflows. This comparison also fed into the portfolio concentration work in Section 7.

[Insert AUM Growth Chart]

5.3 Investor Analytics

On the investor side, the data was broken down by demographics, SIP behaviour, geography and transaction type. A few patterns stood out immediately: SIP transactions were far more consistent month-to-month than lumpsum investments, investors in the 25–40 age bracket made up the largest share of new folios opened during the period, and metro cities accounted for a larger share of transaction value than of transaction count — suggesting investors outside metros tend to invest smaller amounts more frequently.

A large rectangular area with a light blue background and a dashed blue border, intended for charts.

[Insert Investor Analysis Charts]

5.4 What the EDA Changed

This exploratory pass changed two things about the rest of the project. First, it confirmed that risk-adjusted metrics (Section 6) would be more useful than raw returns, since two funds with similar CAGR sometimes had very different volatility profiles. Second, it justified building an investor cohort and SIP continuity analysis (Section 7) — the raw transaction data hinted at a real difference in behaviour between long-term SIP investors and one-off lumpsum investors, and that difference was worth quantifying properly.

6. Fund Performance Analytics

Once the cleaned data was in place, the next step was to calculate a standard set of performance metrics for every scheme in the dataset. These are the numbers that most investors and analysts already look at, so getting them right — and computing them consistently across all schemes — was the first real analytics milestone in the project.

Metric	What It Measures
CAGR (Compound Annual Growth Rate)	Measures the smoothed annual growth rate of a fund over a chosen period. Useful for comparing long-term growth across funds with different holding periods.
Sharpe Ratio	Return earned per unit of total risk taken, using standard deviation as the risk measure. A higher Sharpe ratio means the fund is generating better returns for the volatility it carries.
Sortino Ratio	Similar to Sharpe, but only penalises downside volatility. Two funds with the same Sharpe ratio can have very different Sortino ratios if one of them has more upside volatility than downside.
Alpha	The extra return a fund generates compared to its benchmark, after adjusting for the risk taken. Positive alpha suggests the fund manager added value beyond what the market alone would explain.
Beta	A measure of how sensitive a fund is to overall market movement. A beta above 1 means the fund tends to move more than the market; below 1 means it tends to move less.
Maximum Drawdown	The largest peak-to-trough decline in NAV over the period studied. This is a practical way to see how painful the worst period for a fund actually was, regardless of its average return.

6.1 Building the Fund Scorecard

Individual metrics are useful, but they don't tell the whole story on their own — a fund with a great Sharpe ratio might still carry a high expense ratio that eats into net returns. To address this, a fund scorecard was built by combining return, risk and expense-related factors into a single weighted score.

The scorecard gives each fund a normalised score out of 100, built from four components: risk-adjusted return (Sharpe and Sortino, weighted most heavily), long-term growth (CAGR), downside protection (Maximum Drawdown), and cost efficiency (expense ratio relative to category average). This weighting can be adjusted, but the default setup favours funds that combine steady growth with genuine downside control rather than funds that simply outperform on returns alone.

[Insert Fund Scorecard Table / Ranking Visual]

6.2 Category-Level Observations

Running these metrics across the full dataset surfaced a few clear patterns. Large-cap equity funds tended to cluster together on both CAGR and Sharpe ratio, which makes sense given how closely they track broad market indices. Mid-cap and small-cap funds showed a much wider spread — a handful of standout performers pulled the category average up, while several funds lagged noticeably on risk-adjusted return despite reasonable headline CAGR. Debt funds, as expected, showed low beta and low drawdown across the board, but also the lowest CAGR of any category, confirming their role as a stability anchor rather than a growth driver in a portfolio.

7. Advanced Analytics

Standard performance metrics answer 'how has this fund done.' This section goes a step further and asks 'how much could an investor actually lose, and how are investors really behaving over time.' These techniques are less commonly covered in introductory fund analysis, so each one is explained briefly before describing how it was applied here.

7.1 Historical Value at Risk (VaR)

VaR estimates the possible downside for a fund over a given period at a chosen confidence level. This project calculated 95% confidence historical VaR for each scheme, meaning the loss on any given day should not exceed the VaR figure roughly 95% of the time, based purely on historical NAV movement. This gives a more concrete, day-to-day risk figure than standard deviation alone.

7.2 Conditional VaR (CVaR)

CVaR goes one step further than VaR by measuring the average loss on the days that fall beyond the VaR threshold — in other words, it answers 'when things do go wrong, how bad does it usually get?' This matters because two funds can have the same VaR but very different tail behaviour; CVaR captures that difference.

7.3 Rolling Sharpe Ratio

Rather than calculating a single Sharpe ratio for the whole period, a rolling Sharpe ratio was tracked using a moving window across the dataset. This shows whether a fund's risk-adjusted performance has been improving, deteriorating, or staying roughly constant — a single-point Sharpe ratio can hide a fund that used to perform well but has been sliding for the last year.

7.4 Investor Cohort Analysis

Investors were grouped by the year of their first investment, creating cohorts that could be tracked over time — how much of each cohort was still actively investing a year later, two years later, and so on. This is a common technique in product analytics, and it translates well here: it shows whether newer investors are sticking with SIPs at the same rate as investors who joined a few years ago.

7.5 SIP Continuity Analysis

This analysis specifically flagged investors whose SIP payments had become irregular — missed instalments, inconsistent amounts, or long gaps between contributions. Identifying this group separately from active SIP investors makes it possible to see how much of the platform's SIP book is genuinely stable versus at risk of lapsing.

7.6 Fund Recommendation System

A simple, rules-based recommendation approach was built to suggest funds based on an investor's stated risk appetite (conservative, moderate, or aggressive). It works by filtering the fund scorecard from Section 6 against category and risk-metric thresholds appropriate to each risk profile, then ranking the shortlist by scorecard value. This is not a machine learning model — it is a transparent, rules-based filter, which was a deliberate choice so the reasoning behind any recommendation stays easy to explain.

7.7 Portfolio Concentration (HHI)

The Herfindahl-Hirschman Index (HHI) was used to measure how concentrated AUM is across fund houses and across categories. A high HHI signals that a small number of players dominate a segment, while a low HHI signals a more evenly spread market. This was applied both at the industry level (how concentrated is AUM across fund houses) and at the category level (how concentrated is AUM within, say, small-cap funds specifically).

[Insert VaR / CVaR and Cohort Retention Charts]

8. Power BI Dashboard

The final piece of the platform is a four-page interactive Power BI report connected directly to the SQLite database. It was designed so that a reviewer with no data background can explore the numbers on their own — filtering by category, fund house, time period, or investor segment without needing to write a single query.

8.1 Page 1 — Industry Overview

- Total AUM across the industry, with month-on-month change
- SIP inflows for the selected period
- Total folio count
- Total scheme count across fund houses
- AUM trend line over the full 2022–2026 window

[Insert Screenshot — Industry Overview]

8.2 Page 2 — Fund Performance

- Return-versus-risk scatter plot, positioning every fund by CAGR against Sharpe ratio
- The fund scorecard from Section 6, sortable by any component
- NAV comparison view for selected schemes
- Filters by category and fund house

[Insert Screenshot — Fund Performance]

8.3 Page 3 — Investor Analytics

- State-wise investment map showing geographic distribution of AUM
- Breakdown of SIP, lumpsum and redemption activity
- Age-group analysis of active investors

[Insert Screenshot — Investor Analytics]

8.4 Page 4 — SIP & Market Trends

- SIP inflow trend over time, split by category
- Market comparison view against a benchmark index
- Category-wise inflow analysis showing where new money is going

[Insert Screenshot — SIP & Market Trends]

8.5 Interactivity

Every page shares a common set of slicers for date range, fund category and fund house, so a filter applied on one page carries through to the others. Drill-through was set up from the Industry Overview page down to Fund Performance, so a reviewer can click on a fund house's AUM bar and land directly on a filtered performance view for that fund house's schemes. This kind of cross-page interactivity was one of the more time-consuming parts of the build, but it is also what makes the dashboard feel like a genuine exploration tool rather than a set of static charts.

9. Key Findings

Pulling together the analysis from the previous sections, a set of clear findings emerged. These are summarised below, with a short explanation of what each one means in practice.

10. Large-cap funds showed comparatively stable performance during volatile periods. Their close tracking of broad market indices meant drawdowns were shallower and recoveries faster than mid-cap or small-cap peers.
11. Risk-adjusted metrics provided a far better basis for comparison than raw returns alone. Several funds with attractive headline CAGR figures scored noticeably lower once Sharpe and Sortino ratios were factored in.
12. SIP investment trends showed consistent, steady growth across the period studied, in contrast to lumpsum investments which were far more sensitive to short-term market sentiment.
13. Investor behaviour varied meaningfully across age groups and locations — younger investors leaned more heavily on SIPs, while metro-based investors contributed a larger share of total transaction value despite a smaller share of transaction count.
14. Diversified portfolios generally showed lower concentration risk, as measured by the HHI analysis, and tended to show smoother AUM growth over time compared to portfolios concentrated in a single category.
15. Some funds experienced noticeably higher downside risk during market corrections than their category average, which was only visible once VaR and CVaR were calculated — the standard performance metrics alone did not flag these funds.
16. Major fund houses contributed a disproportionately large share of overall AUM growth, reinforcing the concentration pattern seen in the industry-level HHI figures.
17. The rules-based recommendation system was effective at matching funds to stated investor risk appetite, and its transparent logic made it easy to explain why a particular fund was suggested — an important property for anything investor-facing.

9.1 Reading These Findings Together

None of these findings is surprising in isolation — most are consistent with general knowledge about how mutual funds behave. What this project adds is that each finding is backed by a specific, reproducible calculation against a real dataset, rather than a general assumption. That is really the point of the whole exercise: turning intuition about fund behaviour into numbers that can be checked, filtered, and drilled into on the dashboard.

10. Conclusion

This project set out to build a complete mutual fund analytics platform, and it ended up covering the full path from raw data to finished dashboard — data engineering, financial analytics, and visualisation, all working off the same underlying database.

The result is a solution that takes raw, messy financial data and converts it into insights that can genuinely support investment analysis and decision-making. The database design keeps the pipeline maintainable, the performance and risk metrics give a rounded view of how funds actually behave, and the Power BI dashboard makes all of it accessible to someone without a technical background.

A few limitations are worth noting honestly. The dataset used here covers a fixed historical window rather than live data, the recommendation system is rules-based rather than machine-learning-driven, and some of the investor demographic fields had to be estimated where source data was incomplete. None of these limitations undermine the core findings, but they do shape where the project could go next.

10.1 Future Improvements

- Real-time data pipelines — connecting the ETL process to a live feed rather than periodic CSV exports, so the dashboard reflects same-day NAV and transaction activity.
- Machine learning-based predictions — moving beyond the current rules-based recommendation system toward a model that learns from actual investor outcomes over time.
- Automated reporting — scheduled generation of the fund scorecard and key findings as a recurring report, rather than a manual export.
- Expanded benchmark comparison — incorporating a wider set of benchmark indices for a more granular Alpha and Beta calculation across sub-categories.

Overall, the project demonstrates that a small, well-structured analytics pipeline can do a lot with mutual fund data — not by using anything exotic, but by being disciplined about data quality, choosing metrics that actually matter to an investor, and presenting the results in a way that is easy to explore.

— End of Report —